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Interpretable COVID-19 chest X-ray detection based on handcrafted feature analysis and sequential neural network

Rukundo Prince [a], Zhendong Niu [a,*], Zahid Younas Khan [b], James Chambua [c], Abdallah Yousif [d], Niyishaka Patrick [e], Batamuliza Jennifer [f]

[a] Beijing institute of technology, China
[b] University of Azad Jammu and Kashmir, Pakistan
[c] University of Dar es Salaam, Tanzania
[d] Sudan Technological University, Sudan
[e] University of Hyderabad, India
[f] University of Rwanda, Rwanda

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ABSTRACT

Deep learning methods have significantly improved medical image analysis, particularly in detecting COVID-19 chest X-rays. Nonetheless, these methodologies frequently inhibit some drawbacks, such as limited interpretability, extensive computational resources, and the need for extensive datasets. To tackle these issues, we introduced two novel algorithms: the Dynamic Co-Occurrence Grey Level Matrix (DC-GLM) and the Contextual Adaptation Multiscale Gabor Network (CAMSGNeT). Ground-glass opacity, consolidation, and fibrosis are key indicators of COVID-19 that are effectively captured by DC-GLM, which is designed to adaptively respond to diverse texture sizes and orientations. It emphasizes coarse texture patterns, adeptly catching significant structural alterations in the texture of chest X-rays, enhancing diagnostic precision by documenting the spatial correlations among pixel intensities and facilitating the detection of both significant and minor irregularities. To enhance coarse feature extraction, we introduced CAMSGNeT, which emphasizes fine features via Contextual Adaptive Diffusion. In contrast to conventional multiscale Gabor filtering, CAMSGNeT improves feature extraction by modifying the diffusion process according to both gradients and local texture complexity. The Contextual Adaptation Diffusion approach adjusts the diffusion coefficient by incorporating both gradient and local variance, enabling intricate texture areas to preserve finer details while smoothing regions are diffused to decrease noise. Air bronchograms and crazy-paving patterns are maintained by this adaptive method, which enhances edge identification and texture characteristics while preserving essential tiny details. Finally, a simple optimized sequential neural network analyzes these refined features, resulting in enhanced classification accuracy. Feature importance analysis improves the model's interpretability by revealing the contributions of individual features to its decisions. Our methodology outperforms numerous state-of-the-art models, achieving 98.27% and 100% accuracy on two datasets, providing a more interpretable, precise, and resource-efficient solution for COVID-19 detection.

1. Introduction

In December 2019, COVID-19, one of the deadliest global pandemics caused by SARS-CoV-2, a zoonotic coronavirus, resurfaced. In March 2024, Worldometer reported an estimated seven hundred million cases of infections worldwide, with almost seven million deaths [1].

It is well known that respiratory droplets, direct touch, or close contact are the primary methods by which diseases are spread [2]. To make sure the virus is prevented from spreading, early diagnosis and detection are needed. In order to detect COVID-19, several models, including RT-PCR, machine learning, and deep learning models, have been presented [3–5].

However, these models have a number of significant drawbacks. According to Panjeta et al. [6], there is a possibility that a small quantity of viral material in samples and issues with sample quality, stability, and consistency are likely to render RT-PCR an inadequate method for detecting COVID-19.

* Corresponding author.
E-mail addresses: rukuprince@gmail.com (R. Prince), zniu@bit.edu.cn (Z. Niu), zyounask@gmail.com (Z.Y. Khan), jchambua@gmail.com (J. Chambua), abdosif33@hotmail.com (A. Yousif), niyishakapatrick@gmail.com (N. Patrick), bajelo2000@yahoo.co.uk (B. Jennifer).

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Compared to alternative approaches, deep learning-based models have been proven to be trustworthy, reliable, and accurate. These techniques are widely used and popular, yet they face several challenges, such as the need for large training datasets, expensive computing devices such as GPUs, more trainable parameters, larger feature vector sizes, interpretability challenges, and longer training and testing durations [7]. We proposed a clear, comprehensible, cohesive, and computationally effective approach to address these issues. Compared to state-of-the-art deep learning models for COVID-19 detection in chest X-rays (CXRs), our method has a smaller feature vector size, runs on an inexpensive CPU configuration, and requires less testing and training time.

The key contribution of this manuscript can be summarized in the following manner:

- We presented the Dynamic Co-Occurrence Grey Level Matrix (DC-GLM), a remarkable enhancement in texture analysis. In contrast to traditional GLCM methods that depend on fixed parameters, DC-GLM dynamically adjusts to various texture scales and orientations, making it suitable for the analysis of intricate chest X-ray images. DC-GLM effectively preserves coarse texture patterns by recording the spatial connections among pixel intensities, allowing it to detect both prominent and nuanced variations successfully. This versatility enables DC-GLM to emphasize critical medical features such as ground-glass opacity, consolidation, and fibrosis—hallmarks of COVID-19. DC-GLM markedly enhances diagnostic precision by improving the capacity to detect both extensive structural alterations and subtle abnormalities.
- To complement coarse feature extraction, we developed CAMSGNeT, which focuses on fine features through Contextual Adaptive Diffusion. In contrast to the traditional Gabor filter, CAMSGNeT enhances feature extraction by modifying the diffusion process according to gradients and local texture complexity. To maintain finer details in complex texture areas while reducing noise in smoother parts via diffusion, the Contextual Adaptive Diffusion approach integrates gradient and local variation to modify the diffusion coefficient. This adaptive method effectively preserves important texture patterns, such as air bronchograms and crazy-paving patterns, enhancing edge detection and texture characterization while maintaining essential tiny details.
- Amidst the widely used deep learning models with millions of trainable parameters, we introduce a sequential neural network architecture that uses a modest number of 181,064 trainable parameters (see Fig. 4). This design choice, coupled with the novel DC-GLM and CAMSGNeT algorithms, significantly reduces both training and testing times and offers practical advantages over more computationally intensive models while maintaining high accuracy.
- Model interpretability: We compute feature importance to elucidate the attributes that affect predictions, thereby enhancing confidence in model outputs and informing feature selection and engineering efforts (refer to Section 5.2). Additionally, the suggested methodology improves comprehension of the problem domain, refines failure analysis by pinpointing error-contributing factors, and enhances communication with non-technical stakeholders (refer to Figs. 11).
- In summary, our proposed model offers the remarkable benefit of facilitating real-time clinical decisions. It achieves a classification accuracy of 98.27% and 100% in COVID-19 detection, with shorter runtimes of 15.67 and 17.72 min for two distinct datasets, respectively (see Table 1).

The remaining portions of the manuscript are presented as follows: Relevant literature is listed in Section 2, and methods used are described in Section 3. The experimental results are discussed in depth in Sections 4 and 5. Sections 6 and 7 finally assess the comparative results and draw conclusions.

2. Related works

Riahi et al. [8] proposed a deep-learning model that uses a three-dimensional representation of the data and applies a three-dimensional CNN architecture. The study broke down the original image into image components using the empirical mode decomposition method in two dimensions. It subsequently combined those image components as a sequence of video frames and assigned that sequence as an input to the 3D Convolutional Neural Network model that was responsible for OHRIAb detection and classification. The 3DCNN model was based on a standard 3D VGG-16 architecture approach modified to integrate a context-aware attention module, which was incorporated with normal fully connected layers for classification purposes. They reported an accuracy of 0.999. The authors, Chorney et al. [9], provided a streamlined microsmodel named AttentionCovidNet, which concentrated on using the Channel Attention Convolutional Neural Network on ECG to detect COVID-19. The authors demonstrated that their model outperforms many existing models with an accuracy of 0.993, a precision of 0.997, a recall of 0.993, and an F1-score of 0.995. These findings show that their approach can be as efficient as standard testing and grow the ability of the ECG to be a diagnostic sign of the disease. As shown by Ju et al. [4], CodeNet utilizes contrastive learning for effective use of the latent image information to depict features and extends the algorithm to modes of diverse data domains. In the course of the competition, CodeNet yielded the best results among all the models its developers used, with an accuracy of 0.942 on the testing dataset.

Xu et al. [10] introduced a dual-stream network utilizing EfficientNet architecture for COVID-19 diagnosis from CT scans. This network integrates crucial information from both spatial and frequency domains of CT scans. In addition, they used adversarial propagation (AdvProp) technology to solve the common problem of insufficient training data in computer-aided diagnosis based on deep learning as well as the problem of overfitting. A feature pyramid network (FPN) was used to merge the features from the two data streams. This model reported both 0.9870 and 0.9958 accuracies for multi-class classification and binary classification tasks, respectively. Zheng et al. [11] investigated the Open-Set Single-Domain Generalization for Medical Image Diagnosis (OSSDG-MID), aiming to train a model on a single source domain to classify samples from the target domain accurately. Their method, the Multiple Cross-Matching Method (MCM), improved the identification of 'unknown' categories by generating auxiliary samples outside the source domain's category space. Experimental results demonstrated that MCM outperformed existing methods for generalization in one domain and classifying images with open sets. Talukder et al. [12] proposed a study to investigate the use of X-ray images and deep learning algorithms for rapid and accurate identification of COVID-19 patients. Their approach involved enhancing detection accuracy by fine-tuning established transfer learning models with appropriate layers. The experiment used a COVID-19 X-ray dataset containing 2000 images. Impressive accuracy rates were achieved, with EfficientNetB4 reaching 1.0 accuracy. This model exhibited excellent accuracy in detecting COVID-19 and lung diseases, making it a promising tool for COVID-19 detection. Gu and al. [13] presented an unsupervised domain adaptation technique for analyzing not only the COVID-19 X-rays but also some related images. Unlike the typical unsupervised domain matching methods that fail in conditional class distribution, they used a balanced slice-Wasserstein distance as the metric as an approach to the challenge of transductive domain adaptation. The high performance of their method was demonstrated via several public domain-matching datasets obtained for different COVID-19 cases. The interpretation of multiple data sets as both source and target domains is the main idea of their approach, and it led to the generation of discriminative and domain-invariant representations, which consequently improved data distribution fitting. In their research, Mezina et al. [14] proposed two components of a new deep neural network structure for X-ray chest image analysis. The initial part introduced global feature capture by using an inception architecture, while the second area involved the merging of inception modules and a vision transformer to analyze local features. To address the given X-ray images that involve multiple diseases, they use a modified asymmetric loss function (ASL) for multi-label classification. Instead of evaluating several diseases from the chest X-rays 15 dataset, their research considered only nine types of diseases found in COVID-19 cases.

Issahaku et al. [15] employed a new multi-modal deep learning model for COVID-19 detection that fused chest X-ray images with cough sound features. The variation layer has been incorporated to raise the detection accuracy capacity. Hence, the integration of cough symptoms with X-ray pictures means an important breakthrough in the COVID-19 detection world. The Fast VGG16 model, a new region-based convolutional neural network (FRCNN) addressing its memory restrictions, offered excellent results with an accuracy of 0.9980, an f1 score of 0.9970, and a validation loss of 0.0132. The model was critically evaluated using chest X-ray images and audio clippings from 2059 COVID-19-positive sample chest X-ray scans and cough recordings of 25,000 people, which provided good proof for practice in the medical field. Sahoo et al. [16] proposed a multi-stage end-to-end architecture to classify whether a patient has COVID-19 or not, identify the infected area, and assess the severity of his disease. They presented a multi-stage end-to-end architecture to diagnose any individual Their strategy, being about lung region identification, COVID-19 discrimination, and severity assessment, has better results compared to the content approach, where the prediction confidences in the models are passed to the base models. Using this approach, they created a matrix of mild, moderate, critical, or extreme. The test showed the best representation capability in such tasks as image cutting and classification. The ensemble segmentation model has registered an IoU score of 0.9899 and a Dice score of 0.9949, whereas the forecast infection segmentation model has outperformed other models with a Dice score of 0.9675 and an IoU of 0.9371. The segmentation-based classification model achieved the best accuracy value in terms of 0.9805, and the F1 score is 0.9776.

3. Method

This paper presents two innovative texture analysis techniques: the Dynamic Co-Occurrence Grey Level Matrix (DC-GLM) and the Contextual Adaptation Multiscale Gabor Network (CAMSGNeT). Initially, DC-GLM dynamically adjusts to various textural scales and orientations to examine intricate chest X-ray patterns, including ground-glass opacity, consolidation, and fibrosis, which are critical indications of COVID-19 refer to Algorithm 1. In contrast to traditional DC-GLM methods [17,18] that depend on fixed parameters, DC-GLM's flexibility enables it to identify both coarse and fine texture patterns, hence effectively detecting both minor and significant anomalies in chest X-rays. Secondly, we utilized the Contextual Adaptation Multiscale Gabor Network (CAMSGNeT) to improve fine feature extraction using Contextual Adaptive Diffusion, which modifies the diffusion process based on both gradients and local texture complexity. This innovative diffusion technique modifies the diffusion coefficient by integrating gradient and local variance, enabling intricate textures to preserve smaller details while smoother areas experience diffusion to diminish noise. Both DC-GLM and CAMSGNeT produce high-dimensional feature vectors. We employed principal component analysis (PCA) [19] to reduce processing demands while maintaining 95% of the data variance. Ultimately, to detect COVID-19, we employed a lightweight, sequential neural network model [20] developed in Keras. The model comprises three dense layers and two dropout layers, rendering it computationally efficient relative to deep convolutional models with millions of parameters. The network utilizes the ReLU activation function in the hidden layers and a softmax output layer for binary classification. This streamlined architecture mitigates overfitting and accelerates the training process while preserving high accuracy in the detection of COVID-19 from chest X-rays.

Fig. 1 shows the proposed model flowchart as described above. Sections Section 3.1, 3.2, and 3.3 provide an in-depth explanation of how the model functions.

3.1. Dynamic Co-Occurrence Gray Level Matrix (DC-GLM)

The Dynamic Co-Occurrence Grey Level Matrix (DC-GLM) enhances the conventional Gray-Level Co-Occurrence Matrix (DC-GLM) by providing a more dynamic and adaptive methodology for feature extraction, especially in the context of chest X-rays for COVID-19 detection. Although DC-GLM [21] quantifies the frequency of pixel intensity value pairs at a specified distance and orientation, it is constrained by its static methodology, employing predefined parameters that may inadequately represent the diverse texture patterns found in various regions of an image. This limitation is critical in chest X-rays, where subtle variations like ground-glass opacity or consolidation may present at different scales and orientations, complicating diagnosis. DC-GLM addresses this issue by dynamically adjusting its parameters to suit different texture scales and orientations, thus facilitating the capture of a considerably broader spectrum of textural features. This adaptability renders it especially proficient in detecting the intricate patterns characteristic of COVID-19, such as ground-glass opacity and consolidation, which are frequently overlooked by the static methodology of DC-GLM [22]. Consequently, DC-GLM offers a more nuanced and precise representation of texture patterns in chest X-rays, resulting in enhanced performance and improved diagnostic accuracy in COVID-19 detection relative to conventional DC-GLM. The primary texture features derived from the DC-GLM are listed below:

3.1.1. Contrast

Contrast measures the local variations of pixel intensities; it is computed as follows:

$$\text{Contrast} = \sum_{i,j} (i-j)^2 \cdot \text{DC-GLM}(i,j) \tag{1}$$

High contrast values in the DC-GLM matrix demonstrate the wideness of texture patterns, including lung ground glass opacities and consolidations, which are common with COVID-19. In the DC-GLM formula, i and j signify the dynamically adjusted gray levels of the corresponding pixels. The DC-GLM matrix is made up of elements where each element gives the frequency of the dynamic gray levels i and j that occur jointly, across varying scales and orientations. The term $(i-j)^2$ accounts for a squared difference associated with the pixel intensities (i and j), allowing DC-GLM to highlight texture variations more precisely than static DC-GLM [18]. This dynamic adjustment brings out subtle discrepancies, emphasizing variations crucial for distinguishing COVID-19-affected regions from the surrounding texture environment.

3.1.2. Correlation

Correlation quantifies the linear relationship between the grey levels of neighboring pixels and is computed as follows:

$$\text{Correlation} = \sum_{i,j} \frac{(i-\mu_i)(j-\mu_j) \cdot \text{DC-GLM}(i,j)}{\sigma_i \sigma_j} \tag{2}$$

High correlation values denote areas where grey levels vary systematically, facilitating the identification of patterns potentially linked to COVID-19. The initial label, μ_i, signifies the mean pixel intensity for i within the image, while the subsequent label, μ_j, denotes the mean pixel intensity for j in the image. Ultimately, DC-GLM(i,j) denotes the frequency of concurrent occurrences of the grey levels i and j within the DC-GLM matrix. σ_i and σ_j, the standard deviations, represent the dispersion of pixel intensities around their mean values. This provides crucial information about the variability of pixel intensities in an image, which is important for the texture analysis used in COVID-19 detection from chest X-ray images [23].

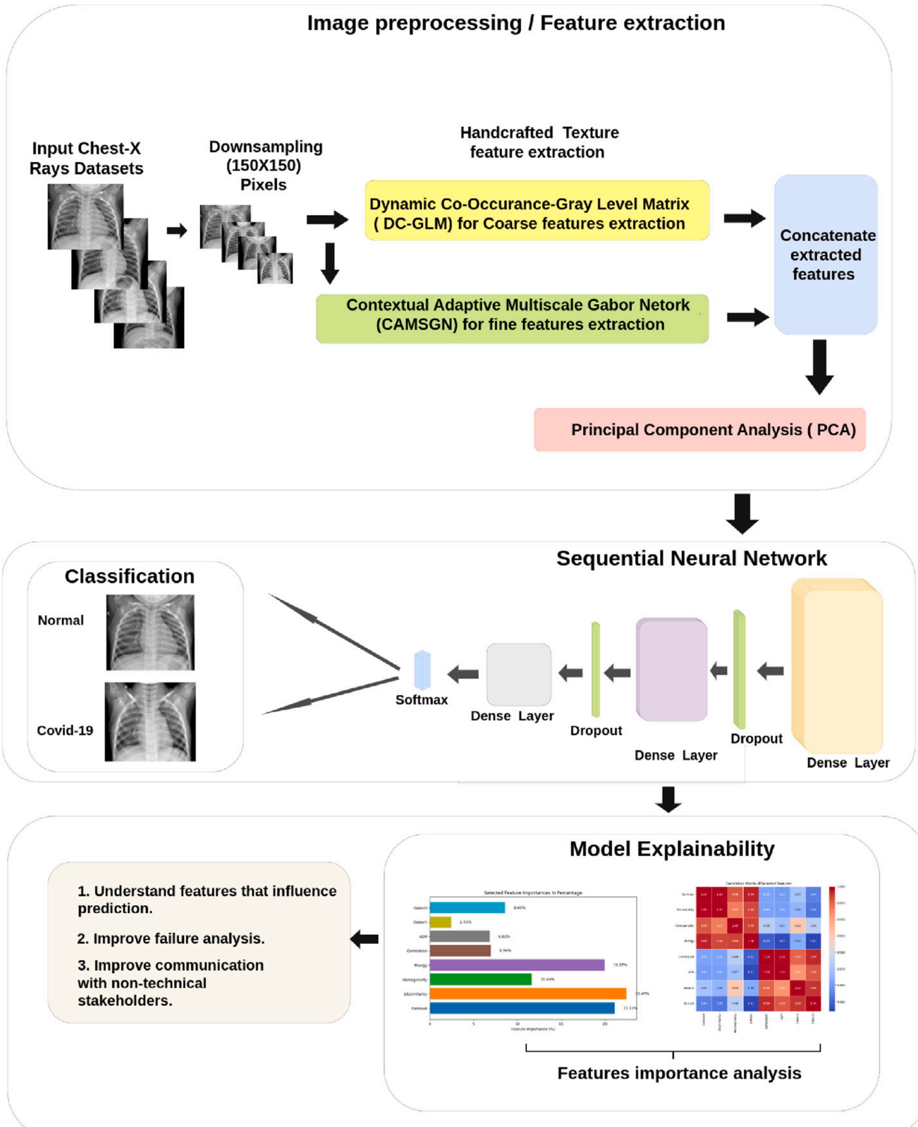

Fig. 1. The proposed system's Flowchart.

3.1.3. Energy

The energy indicates the uniformity or smoothness of the texture.

$$\text{Energy} = \sum_{i,j} \text{DC-GLM}(i,j)^2 \tag{3}$$

This equation denotes the uniformity or smoothness of the texture. Elevated energy values signify regions with more consistent texture patterns, aiding in the detection of COVID-19-related anomalies in normal structures. $\text{DC-GLM}(i,j)^2$ signifies the squared value of an element in the Dynamic Co-Occurrence Gray-Level Matrix (DC-GLM). In texture analysis for COVID-19 detection, this squared value is crucial for quantifying the uniformity or smoothness of texture patterns within the image, providing a more dynamic and adaptive approach for capturing these patterns across various texture scales and orientations compared to the conventional DC-GLM [24]. Fig. 2 illustrates multiple original CXR images alongside their respective DC-GLM-energy variation values.

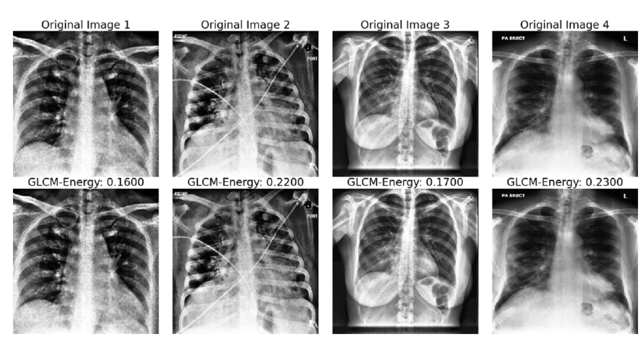

Fig. 2. DC-GLM Energy'texture feature variation across different image processing methods.

3.1.4. Homogeneity

The homogeneity shows the proximity of the distribution of the elements in the DC-GLM to the DC-GLM diagonal and is calculated as follows:

$$\text{Homogeneity} = \sum_{i,j} \frac{\text{DC-GLM}(i,j)}{1+|i-j|} \tag{4}$$

This equation shows how close the distribution of the elements in the DC-GLM is to the DC-GLM diagonal. High homogeneity values indicate regions with more uniform texture patterns and help identify features like consolidations or ground-glass opacities seen in COVID-19.

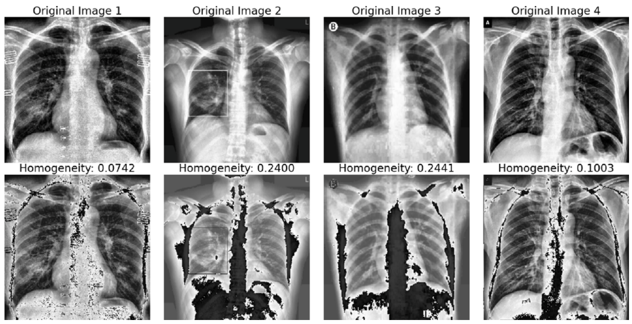

Fig. 3. DC-GLM Homogeneity'texture feature variation across different image processing methods.

By calculating these texture features from the DC-GLM, subtle texture patterns indicative of COVID-19 are captured with greater adaptability to varying scales and orientations compared to traditional DC-GLM, ensuring more accurate detection and classification of the disease in chest X-rays [25]. Fig. 3 highlights different original CXR images and their corresponding DC-GLM-Homogeneity variation value.

The Algorithm 1 explores the procedure for extracting texture features from a greyscale image utilizing the Dynamic Co-Occurrence Grey Level Matrix (DC-GLM). The algorithm begins by initializing variables and calculating the DC-GLM for each pixel in the image at different distances and angles. The matrix delineates the spatial relationships among pixel intensities. Upon constructing the matrix, essential texture features including Contrast, Dissimilarity, Homogeneity, Correlation, and Angular Second Moment (ASM) are computed from the DC-GLM. Contrast emphasizes variations in intensity, whereas dissimilarity quantifies differences in pixel intensity. Homogeneity assesses the consistency of texture, while Energy (ASM) indicates the smoothness or uniformity of texture patterns. Correlation measures the linear association between adjacent pixel intensities. These features are crucial for recognizing distinct patterns, such as ground-glass opacities or consolidations, in medical images, including chest X-rays for COVID-19 diagnosis. The extracted features are subsequently concatenated into a singular feature vector, the input for subsequent classification or analysis. This dynamic methodology enhances the adaptability and precision of texture feature extraction, offering more nuanced and detailed information than conventional techniques.

Algorithm 1: Dynamic Co-Occurrence Gray Level Matrix (DC-GLM) Algorithm

Input: Image I
Output: DC-GLM features: contrast, dissimilarity, homogeneity, correlation, Angular Second Moment

Step 1: Preprocessing the Image
Resize the input image I to a uniform size (150x150).
Convert the image to grayscale.
Step 2: Dynamic Texture Scale and Orientation Adjustment
foreach *pixel in the image* **do**
  Dynamically adjust the texture analysis by utilizing varying texture scales ($d = 1, 2, 3$) and orientation angles (0°, 45°, 90°, 135°, 180°, 225°, 270°) to comprehensively capture pixel relationships. These distances ($d = 1, 2, 3$) are crucial as they capture fine-scale details, medium-range patterns, and broader structural textures in the image. The angles ensure that texture features are extracted from all possible orientations, while the varying distances ensure multiscale analysis of texture patterns.
end
Step 3: Construct the Dynamic Co-Occurrence Matrix (DC-GLM)
foreach *pixel pair* (i, j) **do**
  foreach *dynamic scale and orientation* **do**
    Compute the co-occurrence matrix for the dynamically chosen parameters.
    Store the results in the DC-GLM matrix.
  end
end
Step 4: Compute the Texture Features from DC-GLM
foreach *co-occurrence matrix in DC-GLM* **do**
  Contrast:
$$\text{Contrast} = \sum_{i,j} (i-j)^2 \cdot \text{DC-GLM}(i,j)$$
  Dissimilarity:
$$\text{Dissimilarity} = \sum_{i,j} |i-j| \cdot \text{DC-GLM}(i,j)$$
  Homogeneity (Inverse Difference Moment):
$$\text{Homogeneity} = \sum_{i,j} \frac{\text{DC-GLM}(i,j)}{1+|i-j|}$$
  Energy (Angular Second Moment):
$$\text{Energy} = \sum_{i,j} \text{DC-GLM}(i,j)^2$$
  Correlation:
$$\text{Correlation} = \sum_{i,j} \frac{(i-\mu_i)(j-\mu_j) \cdot \text{DC-GLM}(i,j)}{\sigma_i \sigma_j}$$
end
Step 5: Return the Feature Vector
Return the computed feature vector containing all the texture features: contrast, dissimilarity, homogeneity, correlation, and Angular Second Moment.

3.2. Contextual Adaptation Multiscale Gabor Network (CAMSGNeT)

CAMSGNeT presents a novel approach to medical image analysis, combining the benefits of multiscale Gabor filtering with a very versatile diffusion mechanism. The design is particularly beneficial for detecting COVID-19 in chest X-rays, where both fine and coarse texture patterns are crucial key factors. CAMSGNeT was created to rectify the deficiencies of traditional Gabor filters and improve the precision and clarity of medical image processing.

Significance in Medical Imaging Analysis, especially for COVID-19 detection, it is essential to accurately differentiate both tiny characteristics (e.g., air bronchograms) and larger structures (e.g., consolidation patterns). Traditional approaches, often struggle to achieve a balance between noise reduction and feature preservation across various scales. CAMSGNeT resolves this issue by integrating Contextual Adaptive Diffusion, which adjusts to the intricacy of local textures. This guarantees the preservation of complex information in areas with significant variability while minimizing noise in less complicated places. The versatility of CAMSGNeT renders it very appropriate for evaluating chest X-rays, where texture and contrast can fluctuate markedly across various lung areas. Algorithm 2 describes the procedure for extracting multi-scale texture features that capture fine-grained and localized patterns from CXR images.

3.2.1. Key characteristics of CAMSGNeT

Multiscale Gabor Filtering: Gabor filters are recognized for their capacity to detect orientation-specific characteristics at multiple scales. CAMSGNeT enhances this by dynamically modifying the scale and orientation to identify elements ranging from intricate textures to expansive structural patterns. This guarantees thorough feature extraction from medical chest X-ray images.

Contextual Adaptive Diffusion: In contrast to conventional anisotropic diffusion, which solely accounts for gradients, CAMSGNeT integrates local texture complexity (variance) into the diffusion mechanism. This diffusion method enhances control, maintaining essential boundaries and textures in places with important variability, such as those exhibiting pronounced lung abnormalities, while diminishing noise in uniform areas. The model's capacity to adjust its filtering according to the local image context renders it especially proficient at identifying subtle texture alterations evident in the early phases of COVID-19.

Effective Noise Mitigation: Utilizing Contextual Adaptive Diffusion, CAMSGNeT markedly diminishes noise while preserving sharp edges, hence safeguarding critical medical attributes such as ground-glass opacities and consolidations [26] during preprocessing.

3.2.2. Contextual adaptive diffusion

The Contextual Adaptive Diffusion equation is a sophisticated method for image processing by integrating the image gradient and the local variance of pixel intensities to regulate the diffusion process. This method involves the iterative updating of the picture, where the diffusion at each iteration is influenced by both the image gradient and the complexity of the local texture [27,28].

The diffusion coefficient is expressed as $g\left(|\nabla I_t|, \sigma_{\text{local}}^2\right)$, where ∇I_t denotes the image gradient, and σ_{local}^2 indicates the local variance.

$$I_{t+1} = I_t + \gamma \nabla \cdot \left(g\left(|\nabla I_t|, \sigma_{\text{local}}^2\right) \nabla I_t\right) \tag{5}$$

The contextual adaptive diffusion coefficient is defined as follows:

$$g\left(|\nabla I_t|, \sigma_{\text{local}}^2\right) = \exp\left(-\frac{|\nabla I_t|^2}{(k+\sigma_{\text{local}}^2)^2}\right) \tag{6}$$

In this equation:

- I_t is the image at iteration t.
- γ is the diffusion constant.
- ∇I_t is the image gradient.
- $g\left(|\nabla I_t|, \sigma_{\text{local}}^2\right)$ is the diffusion coefficient, which is modulated by both the image gradient and local variance σ_{local}^2.
- k is the edge threshold parameter.

3.2.3. Multiscale gabor filtering

After contextual adaptive diffusion, multiscale Gabor-like filters are utilized. This filter at a specific scale and orientation is indicated as:

$$G(x, y; \lambda, \theta, \psi, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos\left(2\pi \frac{x'}{\lambda} + \psi\right) \tag{7}$$

Where:

$$x' = x\cos(\theta) + y\sin(\theta)$$
$$y' = -x\sin(\theta) + y\cos(\theta)$$

The parameters are defined as:

- λ: wavelength of the sinusoidal factor, controlling the scale of the texture features.
- θ: orientation of the filter, allowing the capture of directional texture features.
- ψ: phase offset of the sinusoidal component.
- σ: standard deviation of the Gaussian envelope, controlling the spatial extent of the filter.
- γ: spatial aspect ratio, determining the ellipticity of the filter.

The response of the image to the Gabor filter is given by:

$$R(x, y; \lambda, \theta, \psi, \sigma, \gamma) = I(x, y) * G(x, y; \lambda, \theta, \psi, \sigma, \gamma) \tag{8}$$

where $*$ denotes the convolution operation between the image $I(x, y)$ and the Gabor filter $G(x, y)$.

Algorithm 2: Contextual Adaptive Multiscale Gabor Network (CAMSGNeT) Algorithm

Input: Image I, Gabor parameters $(\lambda, \theta, \psi, \sigma, \gamma)$, diffusion constant γ, number of diffusion iterations T.
Output: CAMSGNeT feature vector for image I

Step 1: Preprocessing the Image
- Resize the input image I to a uniform size (150×150)
- Convert the image to grayscale if necessary.
Step 2: Apply Contextual Adaptive Diffusion
foreach *iteration* $t = 1$ *to* T **do**
  Compute the image gradient ∇I_t.
  Compute the local variance σ_{local}^2 using a Laplacian filter.
  Compute the diffusion coefficient based on the gradient magnitude and local variance:

$$g\left(|\nabla I_t|, \sigma_{\text{local}}^2\right) = \exp\left(-\frac{|\nabla I_t|^2}{(K+\sigma_{\text{local}}^2)^2}\right)$$

  Update the image I using contextual adaptive diffusion:
  $I_{t+1} = I_t + \gamma \cdot \nabla \cdot \left(g\left(|\nabla I_t|, \sigma_{\text{local}}^2\right) \nabla I_t\right)$
end
Step 3: Multiscale Gabor-like Filtering
foreach *scale* $s = 0.5$ *to* 1.0 **do**
  foreach *orientation* $\theta \in \{0, \pi/6\}$ **do**
    Construct the Gabor filter with scale s and orientation θ:

$$A(x, y; \lambda_s, \theta, \psi, \sigma_s, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma_s^2}\right) \cos\left(2\pi \frac{x'}{\lambda_s} + \psi\right)$$

    Where:
    $x' = x\cos(\theta) + y\sin(\theta)$
    $y' = -x\sin(\theta) + y\cos(\theta)$
    Apply the Gabor filter to the contextually diffused image I_T to extract filtered features:
    $F_{s,\theta} = I_T * A(x, y; \lambda_s, \theta, \psi, \sigma_s, \gamma)$
  end
end
Step 4: Feature Combination
- Collect features from all scales and orientations:

$$F = \{F_{s,\theta} \mid s \in [0.5, 1.0], \theta \in \{0, \pi/6\}\}$$

- Flatten the feature maps into a single feature vector f.
Step 5: Return the Feature Vector
Return the combined feature vector f that represents both fine and coarse textural details.

3.3. Sequential Neural Network

In order to perform classification tasks, the Sequential Neural Network (SNN) is utilized. This network makes use of a concatenated feature vector that combines coarse texture features from DC-GLM with fine-scale features from CAMSGNeT. DC-GLM captures significant coarse texture attributes, contrast, dissimilarity, homogeneity, energy, correlation, and ASM, whereas CAMSGNeT, enhanced by Contextual Adaptive Diffusion, derives multiscale Gabor-like features that adaptively respond to diverse orientations and scales to capture more intricate textures. The Sequential Neural Network includes two hidden dense layers with 128 and 64 units, respectively, both employing ReLU activation. Dropout layers with a rate of 0.5 are utilized to reduce overfitting, while the final output layer, employing softmax activation, categorizes the input as either COVID-19 positive or negative. The model utilizes the Adam optimizer and categorical cross-entropy loss

Model: "sequential_7"

Layer (type)	Output Shape	Param #
dense_21 (Dense)	(None, 128)	51,968
dropout_14 (Dropout)	(None, 128)	0
dense_22 (Dense)	(None, 64)	8,256
dropout_15 (Dropout)	(None, 64)	0
dense_23 (Dense)	(None, 2)	130

Total params: 181,064 (707.29 KB)
Trainable params: 60,354 (235.76 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 120,710 (471.53 KB)

Fig. 4. The proposed sequential model summary.

for effective binary classification.

Algorithm 3: Key Operations in Neural Network Layers

Input: Input x, Weights kernel, Bias bias, Dropout rate p
Output: Final output after applying the layers
3.3.0.1. 1. relu activation function:.

$$\mathrm{ReLU}(x) = \begin{cases} 0, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

3.3.0.2. 2. the output of dense layer:.

$$\text{output} = \text{activation}(\text{input} \times \text{kernel} + \text{bias})$$

3.3.0.3. 3. dropout application:. Generate a binary mask m with probability $1 - p$.
Apply the mask:

$$\text{output} = \text{input} \times \text{mask}$$

3.3.0.4. 4. softmax activation function:. Convert raw scores into probabilities:

$$\mathrm{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

3.3.0.5. 5. binary cross-entropy loss function:. For binary classification, compute the loss as:

$$\text{Binary Crossentropy}(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log(\hat{y}_i) \right.$$

$$\left. + (1 - y_i) \log(1 - \hat{y}_i) \right]$$

The Algorithm 3, describes essential operations in neural network layers.

4. Experimental results

4.1. Adopted platform

We used a Hewlett-Packard EliteBook 820 G3 laptop with a 12.5-inch display, an Intel Core i7-6600U processor, 8 GB of Double Data Rate fourth-generation random access memory, and a 256 GB M.2 solid-state drive. It runs the Ubuntu 23.04 operating system and has Universal Serial Bus Type-C connectivity and Full High-Definition resolution. We divided our dataset into training (70%) and test (30%) sets, presenting comprehensive timings for feature extraction, training, prediction, and feature vector size for each.

4.2. Used datasets

In this manuscript, we focused exclusively on COVID-19 and normal images for our research, sourcing them from two separate Chest X-ray (CXR) datasets in [29], and [30]. The initial dataset consisted of 460, and 1266 COVID-19, and Normal lung CXR images respectively. The second dataset contained 1670, 1672, and 1670 COVID-19 images, normal images, and pneumonia CXR images respectively. Fig. 5 illustrates the distribution of datasets. Fig. 6 presents the snapshot of the second used dataset.

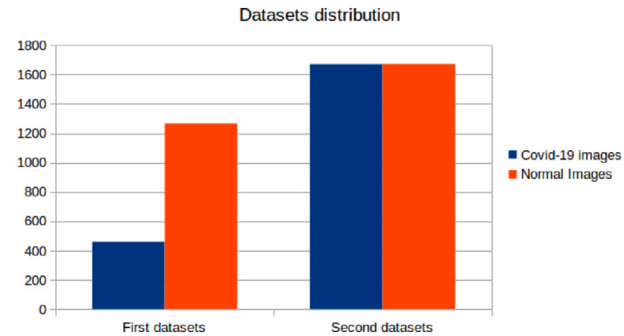

Fig. 5. Dataset description. The first, and second datasets represent [29], and [30] respectively.

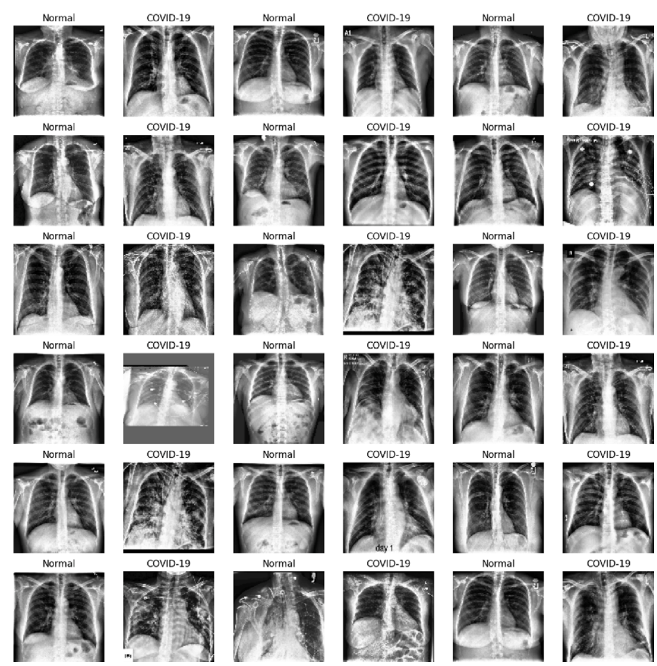

Fig. 6. Snapshot images, from [30] dataset.

A scatter plot illustrating the correlation between the mean and standard deviation of picture pixel values for the "Normal" and "COVID-19" classes is shown in Fig. 7, and Fig. 8. The y-axis displays the standard deviation of pixel intensities throughout the picture channels, while the x-axis displays the mean pixel intensity. The data points are grouped by color, and use unique markers for each class, facilitating the identification of variations in pixel value distributions between the two classes. This graphic illustrates the variances in image texture and pixel intensity between the two classes, providing insights into possible differences in medical imaging properties for diagnostic reasons.

4.3. Metrics

In this study, we evaluate our model using several key metrics: accuracy, recall, precision, specificity, F1-score, and Matthews correlation coefficient (MCC). Accuracy, as shown in Eq. (9), reflects the model's overall prediction correctness [31]. Recall (Eq. (10)) measures sensitivity in detecting true positives, while precision (Eq. (11)) indicates the reliability of positive predictions, reducing false positives [32,

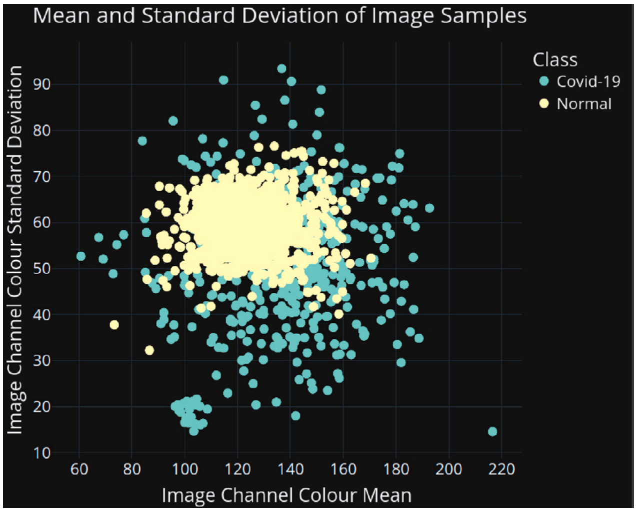

Fig. 7. Mean Value Distribution by Class (Distribution Plot) from [29] dataset.

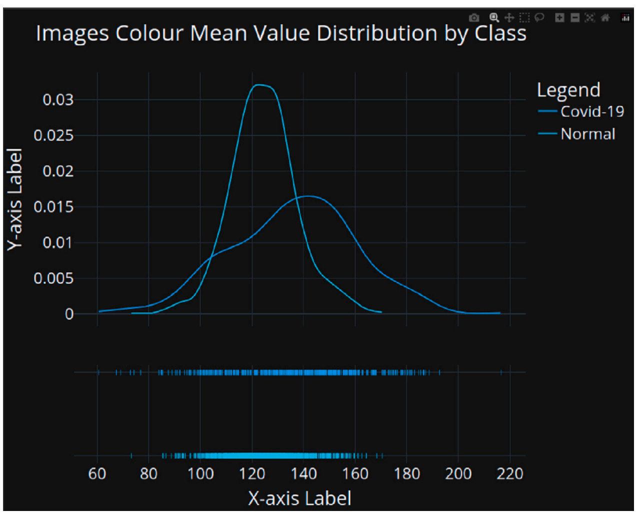

Fig. 8. Mean vs. Standard Deviation Scatter Plot from [29] dataset.

33]. Specificity (Eq. (12)) assesses the model's ability to identify true negatives [34], and the F1-score (Eq. (13)) balances precision and recall [35]. MCC (Eq. (14)) offers a robust evaluation, especially for imbalanced datasets [36]. TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{9}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{10}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{12}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \tag{14}$$

Table 1
The proposed model's training time, testing time, and overall duration in minutes (M) were evaluated using two distinct datasets.

Used dataset	Training time	Testing time	Overall duration
Ullah et al. [29]	15.26	0.41	15.67
El-Sayed et al. [30]	17.15	0.57	17.72

4.4. Running time analysis

Table 1 showcases the result of the model's training hours, and testing duration, with the overall processing time in minutes (M). The first dataset [29] which is 1726 CXR images displayed a running time of 18. 6 min. On the other hand, the second dataset [30] had 3340 CXR images had a running time of 17. 72 min.

5. Results

In this study, we present the results through three key subsections. The classification performance is discussed in Section 5.1, underscores the model's high accuracy, precision, specificity, F1-Score, and MCC, evidencing its robustness and reliability (see Table 2, and Figs. 9–10). Secondly, the feature importance and correlation analysis are discussed in Section 5.2, which investigates the complementary interactions among texture-based, frequency-based, and statistical features, highlighting their role in enhancing diversity and minimizing redundancy (see Fig. 11). Finally, the ablation study is discussed in Section 5.3, rigorously assessing the contribution of individual features and components, validating their significance in achieving the model's optimal performance.

5.1. Classification performance

The proposed model demonstrated robust performance across key evaluation metrics. Table 2 demonstrates a sensitivity of 0.9839 and a specificity of 0.9794, underscoring the models's effectiveness in correctly identifying both positive and negative cases. The precision of 0.9919 highlights the model's high reliability in predicting true positives. Furthermore, the model attained an accuracy of 0.9827 and an F1-Score of 0.9879, indicating its balanced performance across all classes. The Matthews Correlation Coefficient of 0.9574 further validates the model's capability to produce reliable and well-calibrated classifications, reflecting its suitability for real-world applications.

5.2. Features importance, and correlation analysis

Fig. 11 highlights the correlation matrix and features importance. This illustrates the relationships between texture-based features (Contrast, Dissimilarity, and Homogeneity) and frequency-based features (Gabor1 and Gabor2) as well as statistical features (ASM and Correlation). Contrast and Dissimilarity demonstrate moderate correlations with Gabor1 and Gabor2 (ranging from 0.85 to 0.92), indicating that these features contribute complementary, non-redundant information to the model. Furthermore, Homogeneity shows relatively weak correlations with Correlation and ASM tallied (0.88–0.93), underscoring its independent contribution to capturing distinct textural patterns. These findings reflect a balance between diversity and complementarity among the features, which minimizes redundancy while ensuring that the model benefits from a holistic representation of image characteristics. This balance is critical for reducing overfitting and improving the model's generalization capability, particularly in the context of medical image classification tasks.

The feature importance analysis identifies Correlation as the most significant feature (23.18%), reflecting its ability to capture statistical relationships between pixel intensities, which are critical for detecting subtle textural variations indicative of abnormalities in medical images.

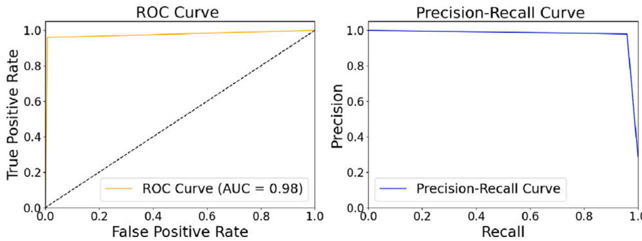

Fig. 9. ROC and PR curve: using Ullah et al. [29] dataset.

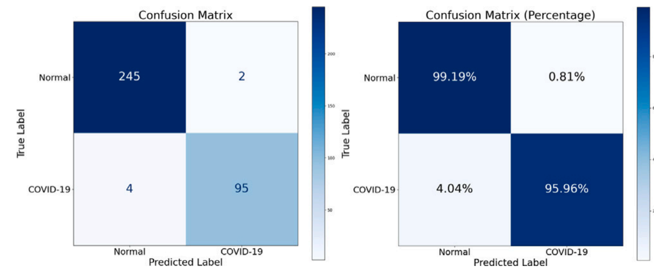

Fig. 10. Confusion matrix and its corresponding percentages.

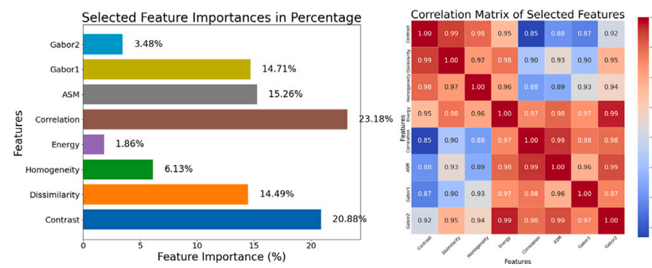

Fig. 11. Feature importance and feature correlation heatmap: using Ullah et al. [29] dataset.

This feature's high importance underscores its role in distinguishing between normal and pathological regions by encapsulating global structural coherence. Conversely, Energy (1.86%) is the least important, as its measure of uniformity or repetitive structures contributes minimally to the classification task, highlighting the model's reliance on more nuanced and discriminative features.

5.3. Ablation study

5.3.1. Omitting CAMSGNeT algorithm

Table 3 highlights the changes in performance metrics across various epochs in the absence of CAMSGNeT. Omitting CAMSGNeT resulted in a notable reduction in precision, specificity, accuracy, and overall diagnostic reliability, as the model relied solely on DC-GLCM features, which lack the intricate frequency-based insights provided by CAMSGNeT. Specifically, CAMSGNeT was designed to extract multi-scale texture features using Gabor-based filters, enabling the capture of fine-grained, localized, and frequency-sensitive patterns that DC-GLCM alone could not achieve. This complementary feature extraction allowed the model to distinguish subtle textural variations in COVID-19-affected and normal lung images, which is essential for accurate and reliable classification. By leveraging contextual and adaptive multi-scale representations, CAMSGNeT enriched the feature set, equipping the model with a more comprehensive understanding of complex image structures and significantly enhancing its generalization across diverse datasets. Its exclusion underscores the critical role of CAMSGNeT in ensuring the model's robustness, interpretability, and diagnostic reliability in medical image analysis.

5.3.2. Computational trade-offs between SNN and CNN

Finally, we replaced the lightweight Sequential Neural Network (SNN) with a more complex Convolutional Neural Network (CNN) model to assess the impact of model architecture on performance. While the CNN model achieved comparable accuracy, it resulted in significantly higher computational demands, both in terms of training time and memory consumption. This increase in resource usage stemmed from the deeper network layers and more extensive parameterization typical of CNNs. Furthermore, the CNN model exhibited a greater tendency to overfit, particularly when trained on the relatively small COVID-19 chest X-ray dataset, highlighting the challenges of applying large-scale, data-hungry architectures to limited medical datasets. In contrast, the SNN, with its lightweight design and fewer parameters, maintained high performance while minimizing computational costs and exhibiting better generalization. This underscores the advantage of the SNN in balancing model complexity with computational efficiency, making it particularly suitable for real-time applications in clinical settings where both performance and resource constraints are critical.

6. Comparative analysis

Our model, as detailed in Table 4, is a computationally efficient, simple framework, as depicted in Table 5, and it utilizes concise parameters, demonstrated in Fig. 4.

Table 2
Performance metrics across various epochs, using the proposed model.

Epochs	Accuracy	Precision	Specificity	F1-Score	MCC
25	0.9798	0.9600	0.9838	0.9648	0.9506
50	0.9769	0.9596	0.9838	0.9596	0.9434
75	0.9827	0.9794	0.9919	0.9694	0.9574
100	0.9798	0.9792	0.9919	0.9641	0.9502
125	0.9827	0.9794	0.9919	0.9694	0.9574
150	0.9827	0.9794	0.9919	0.9694	0.9574
Average	0.9807	0.9728	0.9892	0.9661	0.9527

Table 3
Performance metrics across various epochs, using the proposed model without CAMSGNeT algorithm.

Epochs	Accuracy	Precision	Specificity	F1-Score	MCC
25	0.9538	0.9192	0.9676	0.9192	0.8868
50	0.9595	0.9293	0.9717	0.9293	0.9010
75	0.9566	0.9375	0.9757	0.9231	0.8931
100	0.9682	0.9583	0.9838	0.9436	0.9217
125	0.9653	0.9485	0.9798	0.9388	0.9147
150	0.9653	0.9485	0.9798	0.9388	0.9147
Average	0.9615	0.9402	0.9764	0.9321	0.9053

Table 4 provides an overview of the model, the accuracy achieved, the model's used parameters, and the interpretability information.

Table 5 shows the accuracy and model specifics from various studies. El-Sayed et al. [30] adopted the RESCOVIDTCNNet model with an accuracy of 100%, using EWT, TCN, dilated Causal Convolution Layer, and residual block. Rahman et al. [38] reported an accuracy of 96.29% using ChexNET transfer learning. Chowdhury et al. [39] achieved an accuracy of 99.7% with the DensNet201 transfer learning model.

Our proposed model, contrary, uses a simple design that includes SNN, DC-GLM, CAMSGNeT, and SNN for detecting COVID-19 CXR images. This technique provides an accuracy of 100% while also providing advanced information on model interpretability, a feature missing in the previous studies.

7. Conclusion

In this study, we present a simple, coherent, and easily interpretable model that runs on standard, affordable CPUs, exhibiting less trainable

Table 4
Our proposed model's comparison with Ullah et al. [29], and Sharma et al. [37] focusing on the Model's Specifications, parameters, interpretability, and accuracy.

Adopted model	Obtained accuracy	Model's parameters	Model's interpretability
Ullah et al. [29] (ChestCovidNet)	0.9812	6,417,828	No
Sharma et al. [37] (VGG-16 and Neural Networks)	0.954	138 Millions	No
Our proposed model	**0.9827**	181,064	Yes

Table 5
Comparison of our model with other models that utilized El-Sayed et al.'s dataset [30], with a focus on accuracy, model specifications, and interpretability.

Adopted model	Obtained accuracy	Model's specifications	Model's interpretability
Rahman et al. [38]	0.9629	Convolutional Neural Network for Chest X-ray Image Classification	No
El-Sayed et al. [30]	1	Dilated Causal Convolution Layer + Temporal Convolutional Neural Network + Empirical Wavelet Transform + Residual Block	No
Chowdhury et al. [39]	0.9970	Densely Connected Convolutional Network with 201 layers	No
Proposed model	1	DC-GLM + CAMSGNeT + SNN	Yes

parameters and very limited feature vectors. Despite its simplicity, the proposed model achieves performance levels comparable to those of complex, advanced deep-learning approaches. We adopted textural features extracted from chest X-ray images using the (DC-GLM) Dynamic Co-Occurrence Grey Level Matrix and the Contextual Adaptation Multiscale Gabor Network (CAMSGNeT). A sequential neural network architecture was introduced for analysis and classification tasks. The obtained results denote the potential of a simpler model, providing a transparent alternative to complex deep-learning methods. Finally, we enhanced our grasp of the issue domain, adopting an analysis of feature importance and correlation to detect COVID-19. Our future work will explore the integration of the proposed model with clinical data such as patient demographics and symptoms, to improve COVID-19 detection accuracy and practicality. We will also adopt an external validation, using more diverse datasets to ensure reliability and applicability in clinical settings.

CRediT authorship contribution statement

Rukundo Prince: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Zhendong Niu:** Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Zahid Younas Khan:** Writing – review & editing, Visualization, Validation, Data curation. **James Chambua:** Visualization, Data curation. **Abdallah Yousif:** Visualization, Validation, Resources. **Niyishaka Patrick:** Writing – review & editing, Visualization, Investigation, Formal analysis, Data curation. **Batamuliza Jennifer:** Visualization, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compbiomed.2025.109659.

Data availability

The two used datasets to implement the study are available in the following repositories:

1. CoronaHack -Chest X-Ray-Dataset: https://www.kaggle.com/datasets/praveengovi/coronahack-chest-xraydataset
2. Chest X-ray (Covid-19 & Pneumonia): https://www.kaggle.com/datasets/prashant268/chest-xray-covid19-pneumonia.

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